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GUO-CHENG LI, National Taiwan University, Taipei, Taiwan

POLLY HUANG, National Taiwan University, Taipei, Taiwan

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Measurement and Analysis of a Global-Scale CDN – Locality, Dynamics, and Load Balance

Guo-Cheng Li

NTU

Taipei, Taiwan

r12942085@ntu.edu.tw

Polly Huang

NTU

Taipei, Taiwan

pollyhuang@ntu.edu.tw

Abstract

Content Delivery network (CDN) is a critical part of the Internet service design. It functions as a distributed cache of the content to deliver. Its scale, distribution and caching scheme concerns the user experience that the service providers seek to improve. To facilitate the design and performance analysis process, we measure and analyze the CDN of a popular live video service – Twitch. In that, we conducted a 30-day, global-scale crawl that mapped 2166 edge servers across major continents. The effort, as far as we know, is the most widespread in time-span and geo-diversity in recent years. From the data collected, we observed a two-level hierarchy in the edge server (i.e., the video cache) selection process, similar to that has been reported in [16]. What we see new are 3 salient properties on the locality, dynamics, and load distribution of the servers. (1) The system maintains the viewer-server proximity at the continent granularity. (2) There is a slow rotation of the server clusters over time, and (3) The load distribution is approximately uniform random at both levels of the hierarchy. The tool, methodology, and the findings are important intermediate steps to cost-effective data collection and long-term study of CDNs in operation.

CCS Concepts

• **Networks** → **Network measurement**.

Keywords

CDN measurement, Server selection, Video cache

ACM Reference Format:

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1 Introduction

Content Delivery network (CDN) is key to the user experience of Internet services today. By duplicating or caching content on a large set of servers worldwide, user requests are responded quickly. Video content, in particular, is the most sensitive to the delay and available bandwidth in the delivery path. This makes CDN design critical to the quality of video streaming services. Specifically, the service quality depends on two factors: (1) the size/distribution of

CDN and (2) the server selection strategy. There has been a body of system [1–4] and performance [5–10] studies that co-design the CDN and server selection strategy for performance metrics such as delay, throughput, load balance, and power efficiency. Here, we take a measurement approach and seek lessons from real world designs, much like how [11–13] unveil the CDN and the server selection strategy for global-scale Internet services such as Google, Bing and Akamai. To expand the knowledge base, we report in this work the global CDN and the server selection strategy behind a popular live video service, i.e., Twitch [14]. Discovery of the CDN and its server allocation strategy in Twitch could serve as a reference design for emerging live streaming services or CDN providers. The findings provide also as the performance baseline for studies that seek optimal balances among user experience, system utilization, and/or energy efficiency.

Our contribution is two-fold. We first employ an open-source Twitch CDN crawler, i.e., Kukudy [15]. The crawler was designed to scan as many video channels as possible within Europe [16]. To extend its crawling scope from Europe to the globe and to keep the probing traffic under control, we revise the default crawling strategy such that it is capable of daily discovery of the global CDN. With that, we rotate through 686 VPN nodes and crawl the top 2000 channels daily, from August 15 to September 15 2024. From the 30 daily snapshots, we observe 2166 CDN servers worldwide. In that, 1022, 363, 537, 216, and 28 servers are located in Europe, Asia, North America, South America, and Oceania respectively. This crawl is, as far as we know, the most widespread in time span and geo-diversity in recent years.

Next, we analyze the data collected and discover three salient properties in Twitch’s CDN server selection strategy. (1) A viewer is assigned a server from a set of server clusters in proximity, indicating locality as a factor in Twitch’s server selection strategy. I.e., the viewer and the server clusters are generally close and within the same region. The servers are more often called the *edge servers* for being close to the viewers at the edge of the Internet. (2) Over a long time, one can observe a super server cluster set per viewer. The super set is fixed but the cluster set observed day to day can be different, suggesting dynamics in clusters chosen for service. I.e., the server clusters may rotate in and out of service over time. (3) Similar to Akamai’s server selection strategy [1], the mechanism in Twitch works in two levels. A cluster is selected first from the super set. Within the cluster, an edge server is selected next. While the two-level selection process in Twitch has been observed in [16], we are able to confirm with the new data that the selection is close to uniform random at both levels, indicating load balance as another factor in Twitch’s design. I.e., the CDN operates under a balanced load.



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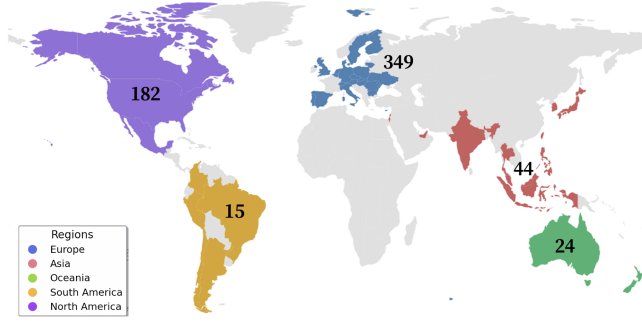


Figure 1: The number of VPN nodes used in the one-month, global-scale crawl by continents.

2 Measurement Methodology

In this study, we employed an enhanced version of Kukudy to explore the global Twitch CDN. This section outlines the strategies for selecting VPN nodes and Twitch channels¹, emphasizing how these decisions ensure a cost-effective collection of the global CDN.

2.1 Crawler

Our crawler was designed to optimize the efficiency of capturing global snapshots of Twitch’s CDN. Utilizing APIs released by Twitch, Kukudy is an efficient crawler that leverages APIs released by Twitch and the VPN provider. It can be configured to request a specific set of channels, receive the corresponding URLs, and discover the servers hosting the channels from worldwide locations, thereby revealing the global server network. Building on Kukudy, we carefully selected VPN nodes and Twitch channels to achieve broad edge server coverage. By targeting a limited yet representative sample of VPN nodes and channels, we obtained a comprehensive view of the CDN while minimizing the time and network resources required.

2.1.1 VPN Node Selection. By using NordVPN [22], we were able to emulate Twitch viewer interactions across different geographical regions. To accurately represent global viewership, we systematically traversed each VPN node to determine its IP address, using the /24 mask to define each unique subnet. In total, **686** unique subnets were identified by this process as of August 13, 2024. Note that NordVPN updates its server infrastructure occasionally, so this number may change over time. Since nodes within the same subnet typically exhibit similar routing and connectivity characteristics, we select only one node in each subnet. Through the mapping of these subnets, we ensure that our crawling system covers Twitch’s global CDN effectively, which improves the representativeness of the data.

As shown in **Fig. 1**, these 686 VPN nodes are distributed across multiple regions, including 349 from Europe, 44 from Asia, 182 from North America, 15 from South America, and 24 from Oceania. Specifically, they cover countries including SE, UK(GB), NL, DE, FR, ES, IT, NO, DK, CZ, PL, AT, FI, BE, CH, BG, CY, HU, LT, MD, PT,

¹These channels are live streams, i.e., mass majority content in Twitch.

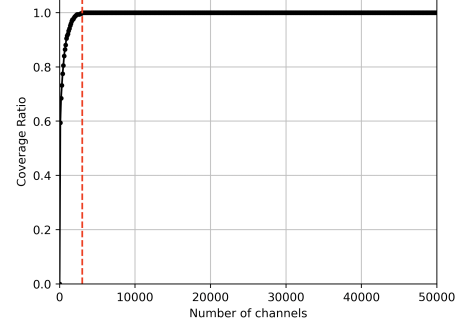


Figure 2: Coverage ratio versus number of channels crawled. Crawling the top 2,000 channels reveals 100% of servers observed crawling all 50,000 channels.

RO, SK, UA, AE, IN, TH, JP, KR, TW, HK, SG, ID, PH, MY, IL, AU, NZ, AR, BR, CL, CO, MX, PE, US, and CA.

It is worth noting that the distribution of available VPN nodes reflects the infrastructure layout of NordVPN, with a higher density in Europe and North America, while regions such as Asia, South America, and Oceania have relatively fewer nodes, especially in remote or less connected areas. Nevertheless, this geographic imbalance does not fundamentally impact the validity of our measurements. The primary goal of our experiment is to observe the CDN server assignment strategy based on viewer geolocation, and our dataset ensures sufficient coverage of major population centers and regional access points. As a result, the observed CDN behavior reflects the practice of server allocation under real-world conditions. Furthermore, as discussed later in Section 3.1.2, we validate that increasing the number of VPN nodes or expanding the set of crawled channels in underrepresented regions results in minimal change to the set of active CDN servers observed.

2.1.2 Channel Selection. The selection of Twitch channels aimed at minimizing the number of channels crawled while maximizing CDN coverage. To explore this, in April 2024 we conducted an experiment using VPN nodes located in different regions, including the United Kingdom, United States, Japan, Australia, and Argentina, to crawl the top 50,000 Twitch channels worldwide.

As shown in **Fig. 2**, CDN coverage increases rapidly with the number of channels crawled. However, after approximately **2,000** channels (indicated by the red dotted line), the coverage ratio approaches 1.0, signifying that most of the edge servers have been reached. Crawling beyond this threshold offers little additional value in terms of coverage, allowing us to focus on 2,000 channels to maximize efficiency.

2.2 Crawling Strategy

Our rate control mechanism takes into account Kukudy’s rate limitations [16] and ensures uninterrupted data collection. Despite Twitch’s maximum request rate of 30 channels per second, we chose to set it slightly lower at 25 channels per second to prevent throttling or interruption during crawling. As for NordVPN’s switching limit of 10 VPN nodes in 4 minutes, we did not need

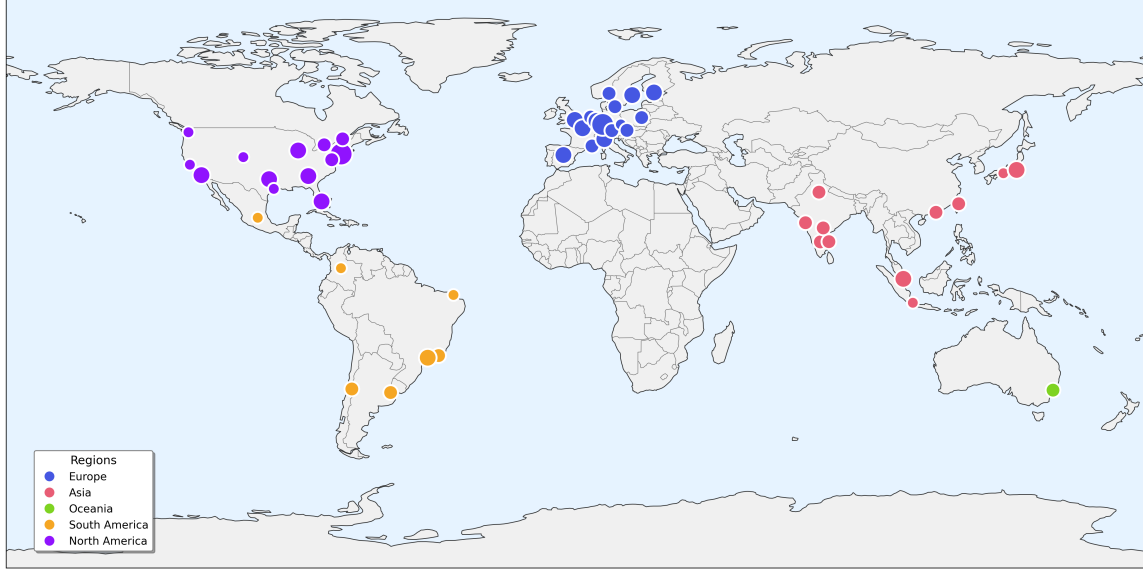


Figure 3: The map of the CDN servers discovered. There are 2166 servers globally, with 1,022, 537, 363, 216, and 28 servers in Europe, North America, Asia, South America, and Oceania. Each circle indicates the location of a server cluster. The size of the circle indicates the number of servers in the cluster in proportion.

to make any adjustments since it takes 80 seconds to crawl 2,000 channels per VPN node far exceeds NordVPN’s switching limit.

For our crawler, crawling 2,000 channels takes 80 seconds, with an additional 20 seconds for connecting to the next VPN node and buffering, resulting in a total crawl time of 100 seconds per node. The buffering time includes the time acquiring the top 2000 channels for the VPN location. Based on 686 VPN nodes, a complete global crawl would require 68,600 seconds, or about **19 hours and 3 minutes**. By monitoring the process of handling channel requests, we avoided detecting other traffic and determined that each channel request requires about 8KB of download traffic and 2KB of upload traffic. With 25 channel requests per second, the total traffic is approximately **200KB/s** downlink and **50KB/s** uplink.

Since peak and off-peak hours differ across countries, sequential traversal of VPN nodes could result in some nodes always requesting channels at the same time of day, limiting the diversity of servers discovered. To mitigate this, we applied uniform random shuffling to the VPN node traversal order prior to each crawl. A new random permutation of the node list is generated daily using a pseudo-random number generator with uniform probability, ensuring equal likelihood of all visiting sequences. This design guarantees that the order varies completely from day to day. Consequently, each VPN node is probabilistically distributed across different time windows over the month-long experiment, covering both peak and off-peak hours.

3 CDN Discovered and Validation

The global crawl of 686 VPN nodes is conducted daily for a month from August 15 to September 15 2024. The results in Section 3, 4, and 6 are the union of the 31 daily CDNs. Section 5 is special. In that, we analyze the difference of the daily CDNs. From the global

crawl, we observed 2166 servers contained in 64 clusters. The server clusters are plotted in Fig. 3. Each circle represents a cluster. The size of the circle reflects the amount of servers in the cluster. Europe is the region with the highest concentration, i.e., 23 clusters with 1,022 servers. North America comes second, hosting 18 clusters with 537 servers. Asia, South America are the 3rd and 4th, hosting 13 clusters with 363 servers and 8 clusters with 216 servers. Oceania hosts only 2 clusters with 28 servers.

3.1 Validation

To assess the data, we compare the CDN discovered by our crawl to the CDN as registered to the authority by Twitch. To assess the crawling strategy, we conducted additional experiments to see whether the result is affected by different crawler settings.

3.1.1 rDNS lookup. We performed rDNS lookups over all IP addresses under AS46489 (Twitch Interactive Inc.) as recorded in IPinfo.io [23]. Only the nodes with a specific hostname pattern are selected. "video-edge-c2b5c8.lax03.abs.hls.ttvnw.net" is the pattern a video server is named in Twitch [21]. The first part of the prefix is the unique host ID and the 2nd part the cluster ID.

Table 1 compares the result from the rDNS lookups and the global crawl. Listed in the table are the specific countries, cities, CDN clusters, the number of registered CDN servers from the rDNS lookups, and the number of CDN servers discovered during the global crawl. According to IPinfo.io, Europe has 29 CDN clusters with 1,793 servers, North America has 25 clusters with 1,153 servers, and Asia has 23 clusters with 850 servers. South America includes 14 clusters with 616 servers, and Oceania has 3 clusters with 70 servers. As compared with other continents, North America and

Table 1: Comparison of the CDN servers reported by IPinfo.io vs. discovered by the global crawl.

Europe					Asia					North America					
Country	City	Cluster	IPinfo	Kukudy	Country	City	Cluster	IPinfo	Kukudy	Country	City	Cluster	IPinfo	Kukudy	
ES	Madrid	mad01	120	71	IN	Bangalore	blr01	23	21	US	Atlanta	atl01	59	51	
		mad02	31	28		Chennai	maa01	31	30		Miami	mia05	64	57	
GB	London	lhr03	45	45	TH	Delhi	del01	23	23	New York	jfk04	104	72		
		lhr04	36	0		Hyderabad	hyd01	26	26		jfk06	72	29		
		lhr08	20	20		Mumbai	bom01	28	26		jfk50	19	14		
		lhr05	1	0		Bangkok	bkk01	32	0		Washington	iad03	19	13	
FR	Paris	cdg02	108	29	ID	Jakarta	bkk02	25	0	Chicago	iad05	69	21		
		cdg10	53	48			jkt01	17	0		ord02	79	0		
NL	Amsterdam	Marseille	mrs02	50	50	CN	Hong Kong	jkt02	18	18	Dallas	ord03	87	45	
		ams02	134	19	hkg01			35	0	ord56		23	19		
		ams03	29	20			hkg06	24	24						
NO	Oslo	osl01	30	25	PH	Manila	mn101	19	0						
IT	Milan	mil02	84	84	SG	Singapore	sin01	54	40		Houston	hou01	21	0	
DK	Copenhagen	cph01	33	30			sin04	42	41			iah50	17	15	
DE	Dusseldorf	dus01	55	53	TW	Taoyuan	tpe01	49	26	Denver	den01	18	0		
	Frankfurt	fra02	193	75			tpe03	33	0		den52	16	11		
		fra05	164	97	JP	Osaka	osa01	18	14	Phoenix	phx01	18	0		
		fra06	94	34		Tokyo	tyo01	17	0	Salt Lake City	slc01	18	0		
	Munich	muc01	29	27		tyo03	41	37	LA	lax03	98	61			
		Berlin	ber01	59	0	tyo05	50	37	San Jose	sjc05	36	0			
CZ	Prague	prg02	1	0	KR	Seoul	sel01	15	0		sjc06	22	16		
		prg03	20	17			sel03	157	0	Seattle	sea01	24	0		
AT	Vienna	vie02	52	50			sel04	73	0			sea02	17	17	
SE	Stockholm	arn03	95	55						CA	Montreal	ymq03	44	40	
		arn04	107	44					Toronto		yto01	55	42		
PL	Warsaw	waw01	30	0											
		waw02	29	29											
FI	Helsinki	hel01	15	0											
		hel03	76	72											
Sum			1793	1022	Sum			850	363	Sum			1153	537	

South America					Oceania						
Country	City	Cluster	IPinfo	Kukudy	Country	City	Cluster	IPinfo	Kukudy		
AR	Buenos Aires	bue01	29	27	AU	Sydney	syd01	29	0		
BR	Fortaleza	for01	28	19			syd02	25	16		
		rio01	51	0			syd03	16	12		
		rio03	126	47							
	São Paulo	rio04	27	0							
sao01		98	0								
sao03		32	23								
sao05		62	48								
scl01		29	21								
CO	Bogotá	bog01	22	16							
MX	Queretaro	qro01	23	0							
		qro02	45	0							
		qro03	18	15							
		qro04	26	0							
Sum			616	216	Sum			70	28		

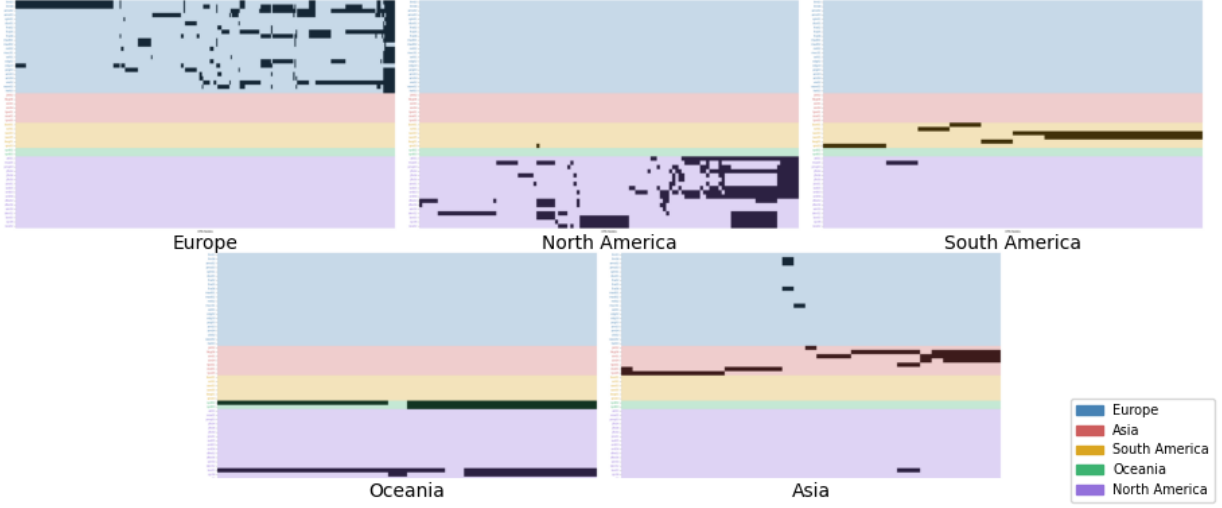


Figure 4: The viewer-server heatmap by the continents. One can see that most viewers are mapped to the clusters in the same continent. There are a few exceptions for the continents that are less resourced.

Europe have a higher number of servers, reflecting greater demand and better infrastructure.

We see generally a higher number of servers and clusters from IPInfo.io. The CDN discovered by our crawl is 57-79% lower in number of clusters and 35-57% lower in number of servers. A similar level of discrepancy has been reported in [16] where Kukudy was used to crawl the CDN in Europe in 2023. In that work, the authors reported seeing 28 clusters and 1755 servers from IPInfo.io and 24 clusters and 1045 servers from an EU-wide crawl. I.e., only a fraction of the servers registered are active. Comparing the result from [16] to the EU part of our data, we see 1 more cluster and 38 more servers registered in 2024. It is not unusual to see an Internet service expanding its CDN infrastructure and adding new clusters to the domain registry. There is however one less cluster and 23 less servers in the EU part of our crawl. This can be attributed to the number of channels crawled in [16]. It is 50 times higher than the number requested in this study. Seeing 2% less servers seems reasonable.

3.1.2 Sensitivity of parameter choices. To assess whether crawling only 2,000 channels and selecting specific VPN nodes would impact the number of clusters and CDN servers discovered, we conducted three experiments targeting countries where significant discrepancies were found between IPInfo.io and our global crawl. They are TH, ID, BR, KR, MX, and PH. In these experiments, we no longer crawled from a subset of VPN nodes but all the VPN nodes in those countries. We increased the number of channels to 20,000 and ensured that each country was covered during peak and off-peak hours. The results showed that expanding the range of VPN nodes and increasing the number of channels did not improve the likelihood of being assigned to the targeted CDN clusters. The number of identified CDN servers remained nearly the same as in the original experiments, with variations of one or two servers

more or less. This suggests that our crawling strategy is efficient in covering the CDN servers that are actively delivering content.

3.2 Discussion

Here we examine further the clusters that we did not find any active servers. For instance, the lhr04, hkg01, sjc05, rio01, and syd01 cluster in **Table 1**. Despite their existence in the IPInfo.io database, these CDN clusters are not active in the period of our crawl. Most of these clusters are one of several clusters serving a city or country. This is not surprising as service providers often adjust their CDNs for maintenance purpose or changes in workload. Less expected are the clusters in Korea, Thailand, and the Philippines. We see no servers in these countries at all. In Korea, IPInfo.io shows that cluster sel01, sel03, and sel04 have a total of 245 IP addresses. However, none of our requests from Korean VPN nodes are assigned to the sel0* clusters. A search on the Web revealed that Twitch clusters in Korea were shutdown in February 2024 due to high network access fee. The Korean clusters are off because of a change in operation cost. Furthermore, compared to other regions, changes in CDN server allocation is more common in Asia, possibly because Twitch is often not the primary choice for video streaming in Asian countries. For example, Japanese and Korean users prefer the domestic streaming platforms, such as Niconico [24] in Japan and AfreecaTV [25] in Korea. Similarly, the service demand in Thailand and the Philippines is likely not substantial or steady enough to justify the cost of operating a local server cluster. Instead, Korean viewers are assigned to CDN clusters in neighboring countries such as Japan or North America, while Thai and Filipino viewers are assigned to CDN clusters in Singapore.

Unlike Korea, Thailand, and the Philippines, other regions did not exhibit such extreme phenomena. The CDN clusters that were not identified are likely backup servers, used for emergency allocation when server load exceeds the capacity, or previously used clusters

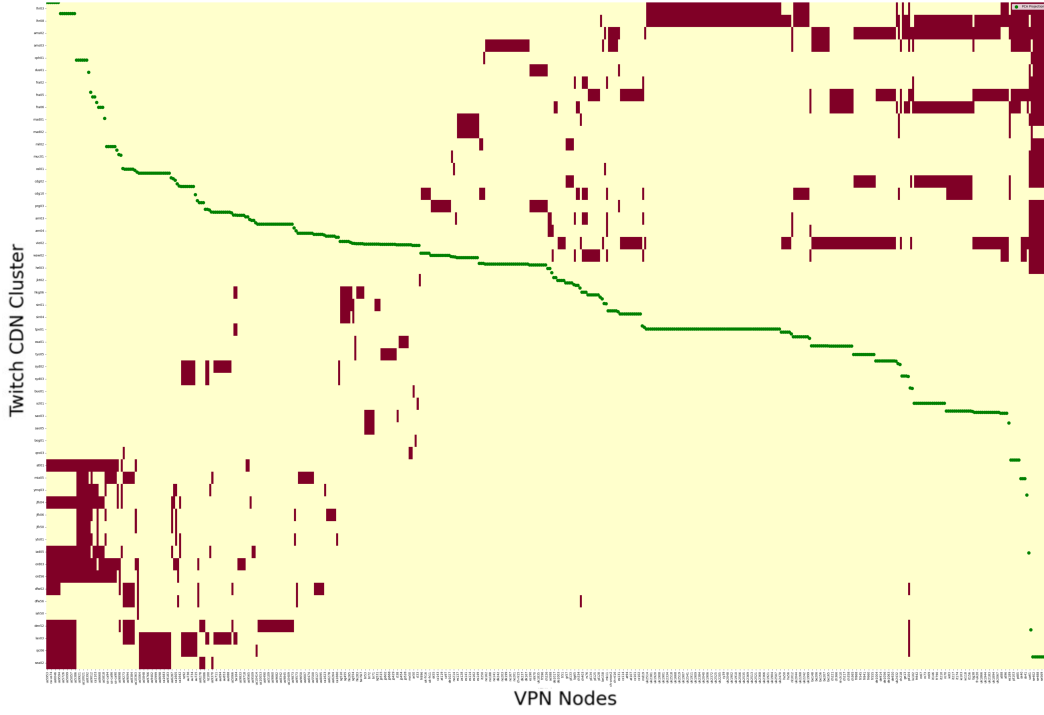


Figure 5: The total viewer-server heatmap sorted by the rank of the principle component.

that are still registered with IPinfo.io. In terms of active servers, Europe shows the highest proportion, with approximately 56% of servers active. This higher percentage could be due to Europe's complex network infrastructure and geographically distributed user base, requiring more active CDN servers to ensure low latency and consistent performance across various countries. North America, Asia, and Oceania have around 40% of their servers active, which represents a more typical balance between active and backup servers. South America shows the lowest activity level, with about 35% of servers active. This lower percentage in South America might be explained by the fact that the users nearby North America can be mapped to North American servers, reducing the need for active servers within South America itself. In the following sections, we will focus our analysis on these **active** CDN clusters.

4 Server Cluster Locality

To show whether the location of the viewers affects server selection, we present first for viewers in a particular continent where the selected server clusters are. Towards a structural understanding, we present the viewer-server relationship next as a heatmap and analyze the relationship by the Principal Component Analysis (PCA).

4.1 Viewer-Server Co-Locality

Fig. 4 depicts where the servers are for viewers in 5 continents. In each of the 5 plots, the x-axis represents individual viewers (i.e., VPN nodes) in the continent and the y-axis gives the location of the servers (i.e., CDN clusters). The servers in a specific continent

are grouped and shown using a unique color. One can see that viewers are typically assigned to a CDN cluster that corresponds to their geographic location, showing that proximity is a factor in the decision. In North America and Europe, viewers are mostly assigned to local CDN clusters, minimizing data transfer distances and optimizing service performance. In South America, a few viewers are directed to North American CDN clusters. This is expected, as South America is geographically close to North America and some viewers in South America may be closer to North American servers.

However, in Oceania and Asia, cross-regional assignments are more common. Oceania has local CDN servers, but most viewers are still assigned to North American CDN servers. The system is likely reassigning viewers to servers in other regions because the local servers cannot handle the full load. In Asia, some viewers are assigned to European servers, particularly in countries like Israel and parts of the Middle East, which are geographically closer to Europe. Like Oceania, some Asian viewers can be assigned to North American servers in case of excessive load. Overall, proximity determines largely how the server cluster is selected, but the mapping can be flexible and adapt dynamically for regions that are under provisioned.

4.2 Heatmap and PCA Analysis

To show the global viewer-server mapping, we visualize the relationship as a heatmap in Fig. 5. The x-axis represents the viewers and the y-axis represents the server clusters. Each grid (x, y) is highlighted when viewer x has been assigned a server in cluster

y in our data. Next, we reduce the dimensionality of the heatmap using PCA. This allows us to capture the primary variations within the dataset. We are able to reveal which viewers behaved similarly in terms of server cluster allocation by comparing the coefficient of the principle component. See the green line in Fig. 5. It indicates the sorted ranking of viewers along the principal component, revealing groups of viewers with similar viewer-server mapping. A visualization such as this shows how certain viewers are more likely to connect to specific servers and can be used to identify anomalies or changes in the server allocation strategy for a long-term analysis of the server allocation policy in the CDN, i.e., a subject of our future work.

5 Server Cluster Dynamics

We explore next whether Twitch adjusts the server cluster assignment dynamically. To this end, we conduct another measurement. The crawling strategy consists of a global crawl once per day. The VPN nodes are traversed in random order each day to ensure that each VPN node (i.e., viewers over the world) requests channels across peak and off-peak hours in a day. We examine the CDN clusters assigned day to day per VPN node. For a given day, we calculate the percentage of CDN clusters reappearing ΔT days later. The metric indicates how much the server cluster set remains the same. Fig. 6 shows the average of such percentage over all VPN nodes. In that, $\Delta T = 1$ compares each VPN node's CDN clusters on day N with $N + 1$. Similarly, $\Delta T = 2$ compares day N with day $N + 2$, and so on. As can be seen, the cluster set shows the highest similarity when $\Delta T = 1$, i.e., compared between two consecutive days. As ΔT increases, the similarity decreases. The decrease in similarity is relatively small, with the difference between $\Delta T = 1$ and $\Delta T = 15$ around 10%. Most of the changes occur in regions with many CDN clusters, such as Europe and North America, where more changes are observed. In contrast, the server clusters in Asia, South America, and Oceania show fewer changes in absolute number. However, if we look at the change in percentage, i.e., the solid blue and red lines in Fig. 6, these regions appear relatively more dynamic in assigning server clusters. This suggests that regions with a large number of CDN clusters tend to adjust frequently in absolute terms, while regions with fewer clusters may appear less dynamic overall but are relatively more volatile when viewed proportionally. Overall, the server clusters do change but slowly, possibly the result of a common practice that rotates the servers in and out of service for maintenance or troubleshooting purpose.

6 Server Load Distribution

We next examine how workload is distributed across servers worldwide, specifically how clusters from the super set and servers within each cluster are selected. While prior work [16] revealed the 2-level hierarchy of this process, the exact load distribution remains unclear. To characterize it, we apply two standard statistical tests: the Empirical Cumulative Distribution Function (ECDF) for uniformity and the Autocorrelation Function (ACF) for randomness. Below we analyze distribution at both the inter-cluster and intra-cluster levels.

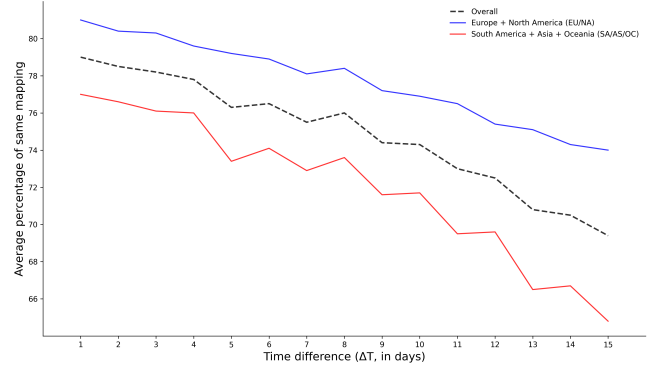


Figure 6: The average percentage of the server clusters remain the same over days (ΔT).

6.1 Inter-cluster Level

The ECDF and ACF plot of server selection process at the intercluster level are illustrated in Fig. 7. On the left, the ECDF plot compares the empirical distribution (represented by the orange solid line) to the ideal uniform distribution (represented by the dashed line). Let N_C represent the number of times cluster C is selected to serve a channel over the 1-month crawling period. The density function (DF) is $P_C = N_C/N$, where N is the total number of channel requests. The cumulative density function (CDF) stacks the density up. In case P_C are equal among all clusters, i.e. a uniform distribution, the DF will be flat and the CDF will be diagonal towards upper right. The cumulative distribution appears to align with the uniform distribution, indicating that most CDN clusters across different regions are utilized fairly evenly. However, in certain regions, the ECDF curve shows sharp increases, revealing that some clusters are allocated significantly more than the others. This happens mainly in Europe and North America, where the number of VPN nodes used in our experiment is higher. On the right, the ACF plot reveals the temporal autocorrelation of cluster usage frequencies. Generally, the autocorrelation values are close to zero, indicating minimal temporal dependency. However, a few points show higher autocorrelation values, indicating that certain clusters may experience recurring usage patterns. This could be due to the clustering of VPN nodes or consistent traffic loads in specific regions. Overall, it shows that Twitch's CDN cluster allocation at the inter-cluster level is largely uniform across regions and independent over time.

6.2 Intra-cluster Level

We observe that intra-cluster distributions are generally consistent across different clusters. Therefore, we select the lhr03 cluster as a representative for discussion. See Fig. 8. Compared to the inter-cluster level analysis, the intra-cluster patterns in the ECDF plot are more distinct, showing a closer alignment with the ideal uniform distribution, suggesting that server utilization within the lhr03 cluster is well-balanced. The ACF plot confirms further that the server assignments are generally independent. Overall, Twitch's server load is balanced across and within clusters with minor variation to adapt to varying traffic loads.

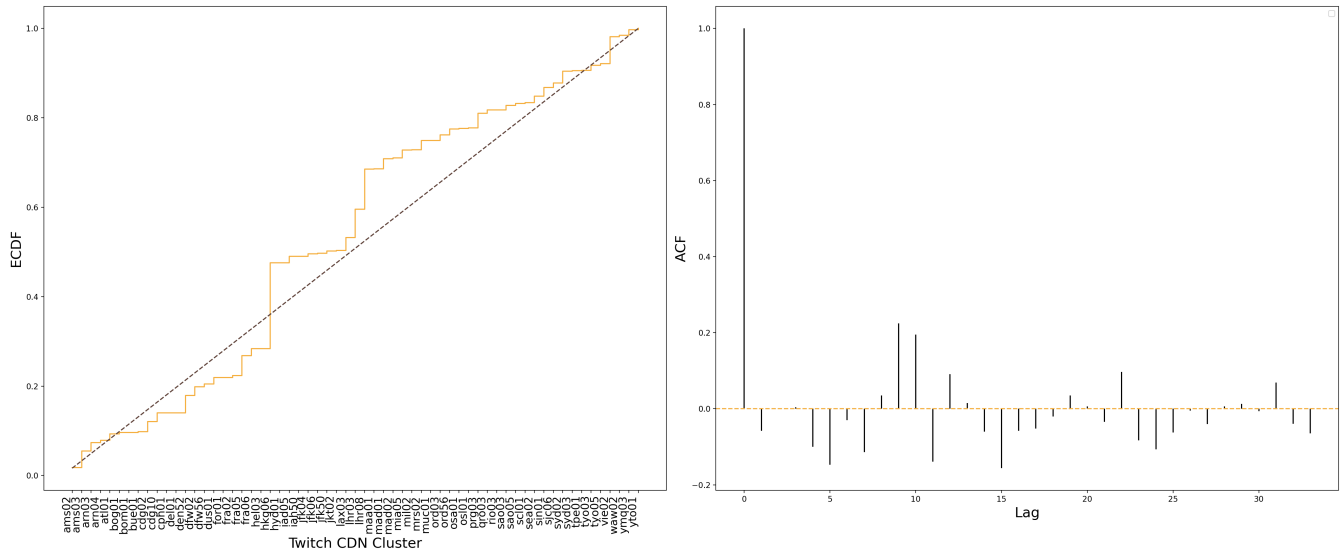


Figure 7: Left: ECDF plot of the server cluster selection process. The server cluster selection process (orange solid line) deviates from a perfect uniform distribution (green dashed line) but not by much. Right: ACF plot of the server cluster selection process. There is no clear autocorrelation in server cluster selection, indicating the process is nearly uniform random.

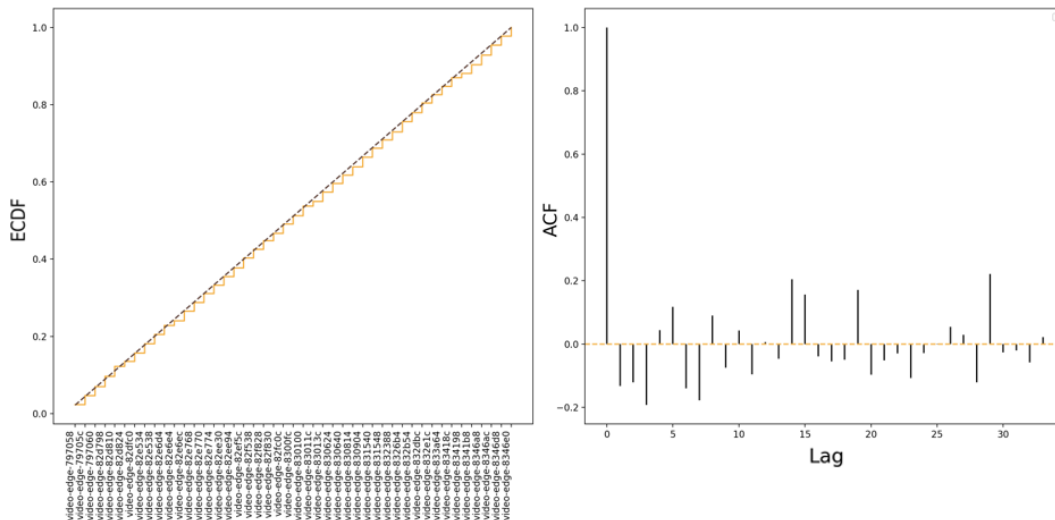


Figure 8: Left: ECDF plot of the server selection process within cluster lhr03. The server selection process (orange solid line) is a perfect fit to a uniform process (green dashed line). Right: ACF plot of the server selection process. There is no clear autocorrelation in the server selection process in cluster lhr03, indicating the process is uniform random.

7 Related Work

Twitch and its CDN are well studied. [17] was the first study unveiling the use of RTMP, HLS, and 3-way redirection along the video distribution pipeline and investigating the arrival process of viewers and the popularity distribution of the videos. [18] presented a dataset by collecting traffic generated from YouTube Live and Twitch and inferred the size of the services in respect of bandwidth usage and the number of concurrent channels. Deng et al. [19] is the first to measure Twitch’s CDN by repeatedly requesting available

streams. To cover a significant geographical span, they exploited 806 proxy servers sending the stream requests worldwide. They were able to observe the set of video servers allocated to viewers from different regions of the world. Their analysis showed that the server allocation policy tend to minimize delay and server load. More recently, Wung et al. [20] performed a partial scan of Twitch’s CDN by deploying a crawler on a cloud computing platform. Wang et al. [21] documented another crawler and their result for deploying it on a VPN platform. Chou et al. [16] devised an efficient

crawler, i.e., Kukudy. The authors scanned Twitch’s CDN in Europe as completely as possible, referred to as a best-effort scan to serve as the best-line to understand the trade-off between CDN coverage and traffic overhead. In this work, we refine the crawling strategy from [16] which enables us to measure Twitch’s global CDN and observe the dynamics of the CDN over a month time.

8 Conclusion

We have achieved a global crawl of Twitch’s CDN and discovered 2166 servers using a lightweight crawling strategy. The crawling strategy facilitates data collection and long-term observation of an Internet-scale eco-system. To consolidate the empirical result, one can compare to data from alternative rDNS services such as Hoiho [26] or data discovered from alternative VPN providers. The data collected give rise to several observations of Twitch’s server selection strategy. The findings facilitate performance analysis of the caching schemes towards cost-effective CDN designs. Towards long-term analysis of large-scale CDNs at work, we seek to explore in the future work the feasibility of a metric such as the principle component ranking proposed in Section 4.2 as a fingerprint of viewer-server mapping, with which one may identify anomalies or structural changes in a large-scale CDN with ease.

A Appendix: Ethical Consideration

With regard to the channels and streamers. Twitch allows both streamers and viewers to download video streams and clips, but reuse without the creator’s consent is a copyright violation. Our crawler stopped at the master playlist server; no video content was downloaded or reused. In addition to public information such as channel ID, title, viewer count, started time, language, game, tags, streamer user ID, user login, and user name, the information collected regarding the channels and streamers was limited to the origin of the stream at a subcontinental level. This origin may suggest the ingest server used, and hence the streamer’s approximate location, but it is coarse enough not to constitute personal or sensitive information.

With regard to Twitch, NordVPN, and the Internet community. Information on active channels was obtained through the official Twitch API, which is commonly used by Twitch users to support analysis and enhance client-viewer interaction. Our crawler interacted only with standard Twitch CDN systems—including the GraphQL endpoint, the Usher, the video weavers, and the video edge servers—all of which are part of the normal data path used by regular viewers. At no point did we exceed the request volume expected of ordinary users: the crawling frequency was kept below Twitch’s published rate limits. Likewise, NordVPN imposes connection limits to prevent abuse, and our measurements respected these restrictions.

References

- [1] Chen, Jiayi, Sharma, Nihal, Khan, Tarannum, Liu, Shu, Chang, Brian, Akella, Aditya, Shakkottai, Sanjay, Sitaraman, Ramesh K: Darwin: Flexible Learning-based CDN Caching. In Proceedings of the ACM SIGCOMM 2023 Conference, pp. 981–999. ACM SIGCOMM ’23, New York, NY, USA (2023)
- [2] Zhou, Mengying and Guo, Tiancheng and Chen, Yang and Wan, Junjie and Wang, Xin: Polygon: a QUIC-based CDN server selection system supporting multiple resource demands. In Proceedings of the 22nd International Middleware Conference: Industrial Track, pp. 16–22. ACM Middleware ’21, Québec city, Canada (2021)
- [3] Mukerjee, Matthew K. and Naylor, David and Jiang, Junchen and Han, Dongsu and Seshan, Srinivasan and Zhang, Hui: Practical, Real-time Centralized Control for CDN-based Live Video Delivery. In Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication, pp. 311–324. ACM SIGCOMM ’15, London, United Kingdom (2015)
- [4] J. Kangasharju, F. Hartanto, M. Reisslein and K. Ross: Distributing Layered Encoded Video through Caches. IEEE Transactions on Computers 51(06), 622–636 (2002)
- [5] Kara, Burak and Simon, Gwendal: Power Efficient Multi-CDN Communication over Content Steering Server. In Proceedings of the 15th ACM Multimedia Systems Conference, pp. 478–484. ACM MMSys ’24, Bari, Italy (2024)
- [6] Usama Naseer and Theophilus A. Benson: Configanator: A Data-driven Approach to Improving CDN Performance. In Proceedings of the 19th USENIX Symposium on Networked Systems Design and Implementation, pp. 1135–1158. USENIX NSDI ’22, Renton, WA (2022)
- [7] Wang, Huan and Tang, Guoming and Wu, Kui and Fan, Jiamin: Speeding up Multi-CDN Content Delivery via Traffic Demand Reshaping. In Proceedings of the 38th IEEE International Conference on Distributed Computing Systems, pp. 422–433. IEEE ICDCS ’18, Vienna, Austria (2018)
- [8] D’Oro, Salvatore and Galluccio, Laura and Palazzo, Sergio and Schembra, Giovanni: A Game Theoretic Approach for Distributed Resource Allocation and Orchestration of Software-defined Networks. IEEE Journal on Selected Areas in Communications 35(3), 721–735 (2017)
- [9] Liu, Chang and Sitaraman, Ramesh K. and Towsley, Don: Go-with-the-winner: Performance based client-side server selection. In Proceedings of the 2016 IFIP Networking Conference (IFIP Networking) and Workshops, pp. 404–412. IEEE IFIP Networking ’16, Vienna, Austria (2016)
- [10] Li, Chenglin and Frossard, Pascal and Xiong, Hongkai and Zou, Junni: Distributed wireless video caching placement for dynamic adaptive streaming. In Proceedings of the 26th International Workshop on Network and Operating Systems Support for Digital Audio and Video. ACM NOSSDAV ’16, Klagenfurt, Austria (2016)
- [11] Calder, Matt and Fan, Xun and Hu, Zi and Katz-Bassett, Ethan and Heidemann, John and Govindan, Ramesh: Mapping the Expansion of Google’s Serving Infrastructure. In Proceedings of the 2013 Conference on Internet Measurement Conference, pp. 313–326. ACM IMC ’13, Barcelona, Spain (2013)
- [12] Calder, Matt and Flavel, Ashley and Katz-Bassett, Ethan and Mahajan, Ratul and Padhye, Jitendra: Analyzing the Performance of an Anycast CDN. In Proceedings of the 2015 Internet Measurement Conference, pp. 531–537. ACM IMC ’15, Tokyo, Japan (2015)
- [13] Chen, Fangfei, Sitaraman, Ramesh K., Torres, Marcelo: End-User Mapping: Next Generation Request Routing for Content Delivery. In Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication, pp. 167–181. ACM SIGCOMM ’15, London, United Kingdom (2015)
- [14] Christine Weber: Twitch State of Engineering 2023 (2023), <https://blog.twitch.tv/en/2023/09/28/twitch-state-of-engineering-2023/>, last accessed 2024/09/14
- [15] Hsuan-Yu Chou: Kukudy, Github (2023), <https://github.com/hy-chou/kukudy>, last accessed 2024/09/14
- [16] Chou, Hsuan-Yu and Huang, Yu-Ting and Huang, Polly: Best-Effort Scan of Twitch’s CDN in Europe. In Proceedings of the 18th Asian Internet Engineering Conference, pp. 10–18. ACM AINTEC ’23, Hanoi, Vietnam (2023)
- [17] Zhang, Cong and Liu, Jiangchuan: On Crowdsourced Interactive Live Streaming: A Twitch.TV-Based Measurement Study. In Proceedings of the 25th ACM Workshop on Network and Operating Systems Support for Digital Audio and Video, pp. 531–537. ACM NOSSDAV ’15, Portland, Oregon (2015)
- [18] Pires, Karine and Simon, Gwendal: YouTube Live and Twitch: A Tour of User-Generated Live Streaming Systems. In Proceedings of the 6th ACM Multimedia Systems Conference, pp. 225–230. ACM MMSys ’15, Portland, Oregon (2015)
- [19] Deng, Jie and Tyson, Gareth and Cuadrado, Félix and Uhlig, Steve: Internet Scale User-Generated Live Video Streaming: The Twitch Case. Amann, J. (eds) Passive and Active Measurement. PAM 2017. Lecture Notes in Computer Science(2017), vol 10176. Springer, Cham.
- [20] Wung, Wei-Shiang and Ting, Guan-Ting and Hsu, Ruey-Tzer and Hsu, Cheng and Tsai, Yu-Chien and Wang, Caleb and Liu, Yuan-Tai and Chen, Hsi and Huang, Polly: Twitch’s CDN as an Open Population Ecosystem. In Proceedings of the 16th Asian Internet Engineering Conference, pp. 56–63. ACM AINTEC ’21, Virtual Event, Japan (2021)
- [21] Wang, Caleb and Liu, Yuan-Tai and Huang, Polly: Juby: Design and Deployment of a Crawler for Twitch CDN Mapping. In Proceedings of the 17th Asian Internet Engineering Conference, pp. 44–52. ACM AINTEC ’22, Hiroshima, Japan (2022)
- [22] NordVPN: NordVPN Website, <https://nordvpn.com/>, last accessed 2024/09/14
- [23] ipinfo.io: AS46489 Twitch Interactive Inc., <https://ipinfo.io/AS46489>, last accessed 2024/09/14
- [24] NicoNico: Japanese Famous Live Streaming Platform, <https://www.nicovideo.jp>, last accessed 2024/09/25
- [25] Afreeca Tv: Korea Famous Live Streaming Platform, <https://www.afreecatv.com/>, last accessed 2024/09/25
- [26] Hoiho: rDNS Lookup Service, <https://hoiho.caida.org/>, last accessed 2025/08/18